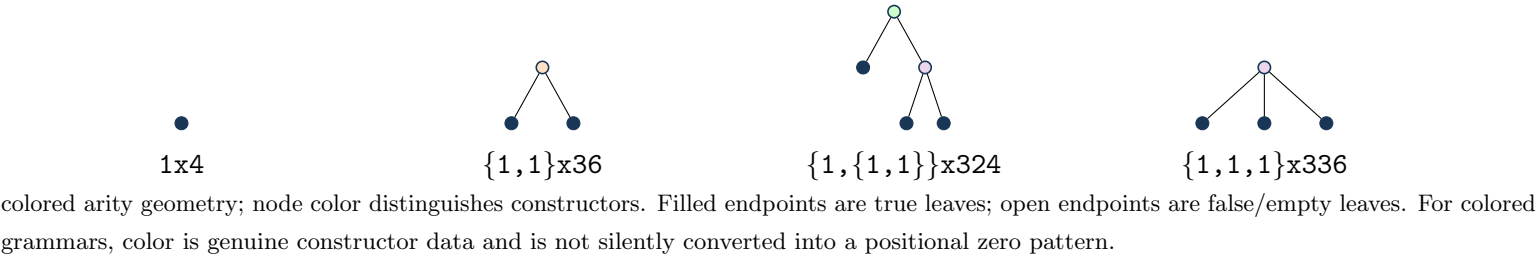


Definition and first terms

$25 A(x) = 24 + 64 x + A(x)^9$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(4)*T(x)$; $T=x+9*T^2+84*T^3+504*T^4+2016*T^5+5376*T^6+9216*T^7+9216*T^8+4096*T^9$.
 $a(0), \dots, a(7) = 1, 4, 36, 984, 31716, 1140552, 43895208, 1768717872$.
 Kernel used throughout: $\rho(u) = u(1 - 9u^1 - 84u^2 - 504u^3 - 2016u^4 - 5376u^5 - 9216u^6 - 9216u^7 - 4096u^8)$, so $x = \rho(T(x))$.

Typogeometry



Typogeometric constructor codes and multiplicities: Δ_2 : 9, Δ_3 : 84, Δ_4 : 504, Δ_5 : 2016, Δ_6 : 5376, Δ_7 : 9216, Δ_8 : 9216, Δ_9 : 4096.

Coefficient and contour extraction

$$\begin{aligned}
 \mathcal{K}_n &= \{(k_1, \dots, k_8) \in \mathbb{Z}_{\geq 0}^8 : \sum_{j=1}^8 j k_j = n - 1\}, \quad K = \sum_{j=1}^8 k_j, \\
 a(n) &= \frac{4}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_8} 9^{k_1} 84^{k_2} 504^{k_3} 2016^{k_4} 5376^{k_5} 9216^{k_6} 9216^{k_7} 4096^{k_8}. \\
 a(n) &= \frac{4}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.
 \end{aligned}$$

Recurrence

$\sum_{r=0}^8 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-8 coefficient polynomials are embedded in `certificate_payload.json` (source: `data/recurrence.json`).

Differential equation

The scalar linear ODE has order 8. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$\begin{aligned}
 R(n, u) &= N(n, u) / \rho(u)^7, \quad \deg_u N = 64. \\
 \text{Telescoping identity: } \sum_r P_r(n) H_{n+r}(u) &= \partial_u (R(n, u) H_n(u)).
 \end{aligned}$$

Matrix dimensions: 18×18 ; remainder 8×9 , rank 8, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.